



Glass Box Fraud Detection

Optimizing Machine Learning
Rigor for Maximum Financial
ROI & Ethical Compliance

Prepared for AIM Academic Panel
& Executive Leadership

THE BUSINESS MANDATE



GOAL

Maximize ROI by balancing intervention and customer friction.



THE THREAT

False Negatives (missed fraud) cause massive direct financial loss.



THE FRICTION

False Positives (false alarms) waste operational resources and alienate good customers.

THE TECHNICAL REALITY



SCALE

1,000,000 total transactions.



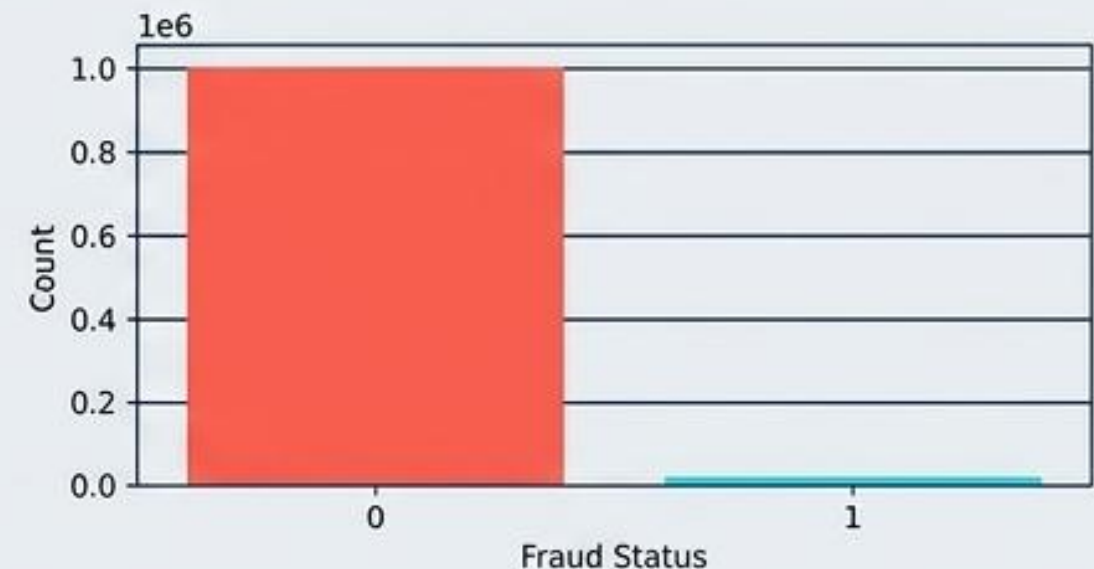
COMPLEXITY

Fraud signals are non-linear and hidden in dense, multidimensional noise.



IMBALANCE

Only 1.1% of transactions are fraudulent.



Feature Engineering Pipeline



Step 1: Foundational Prep

- Audited for nulls and duplicates (none found).
- Identified hidden "-1" missing values and replaced with medians to prevent data leakage.



Step 2: Normalization

- Applied MinMaxScaler to continuous numerical features.
- Calibrated all values to a 0-1 scale to prevent magnitude bias during learning.

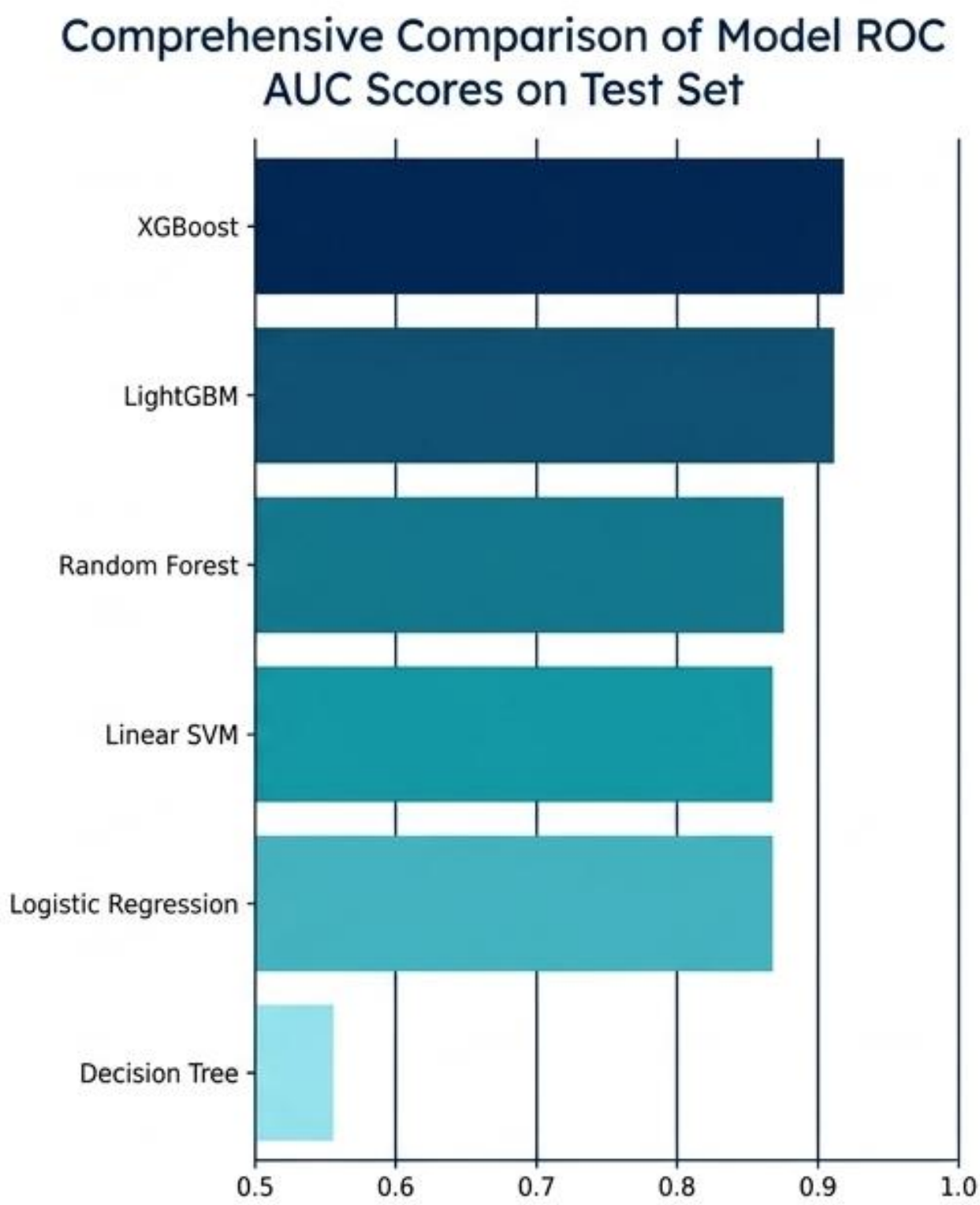


Step 3: Domain-Derived Features

- Engineered complex interaction terms (e.g., income_x_credit_score).
- Calculated behavioral velocity ratios (e.g., velocity_ratio_6h_24h) to capture non-linear fraud signals.

Model Evaluation & Selection Matrix

Diagnostic Comparison Matrix			
Model	Category	Trade-offs	AUC Score
LightGBM (Tuned)	The Winner	Auto-chose col-wise multi-threading for peak efficiency. Achieved 72.6% Recall.	0.832
XGBoost	Top Tier	Strong gradient boosting performance.	0.823
Linear SVC	Contender	Stable, but lacks optimal precision/recall balance.	0.770
Logistic Regression	Baseline	Solid baseline, struggles with non-linear complexity.	0.770
Random Forest	Contender	Stable ensemble, lacks optimization.	0.730
Decision Tree	Underperformer	Fails completely against extreme class imbalance.	0.518



THE FINANCIAL REALITY

Not all classification errors cost the same. Default model thresholds (0.5) are financially reckless.



False Negative (Missed Fraud):
\$1,000 cost.

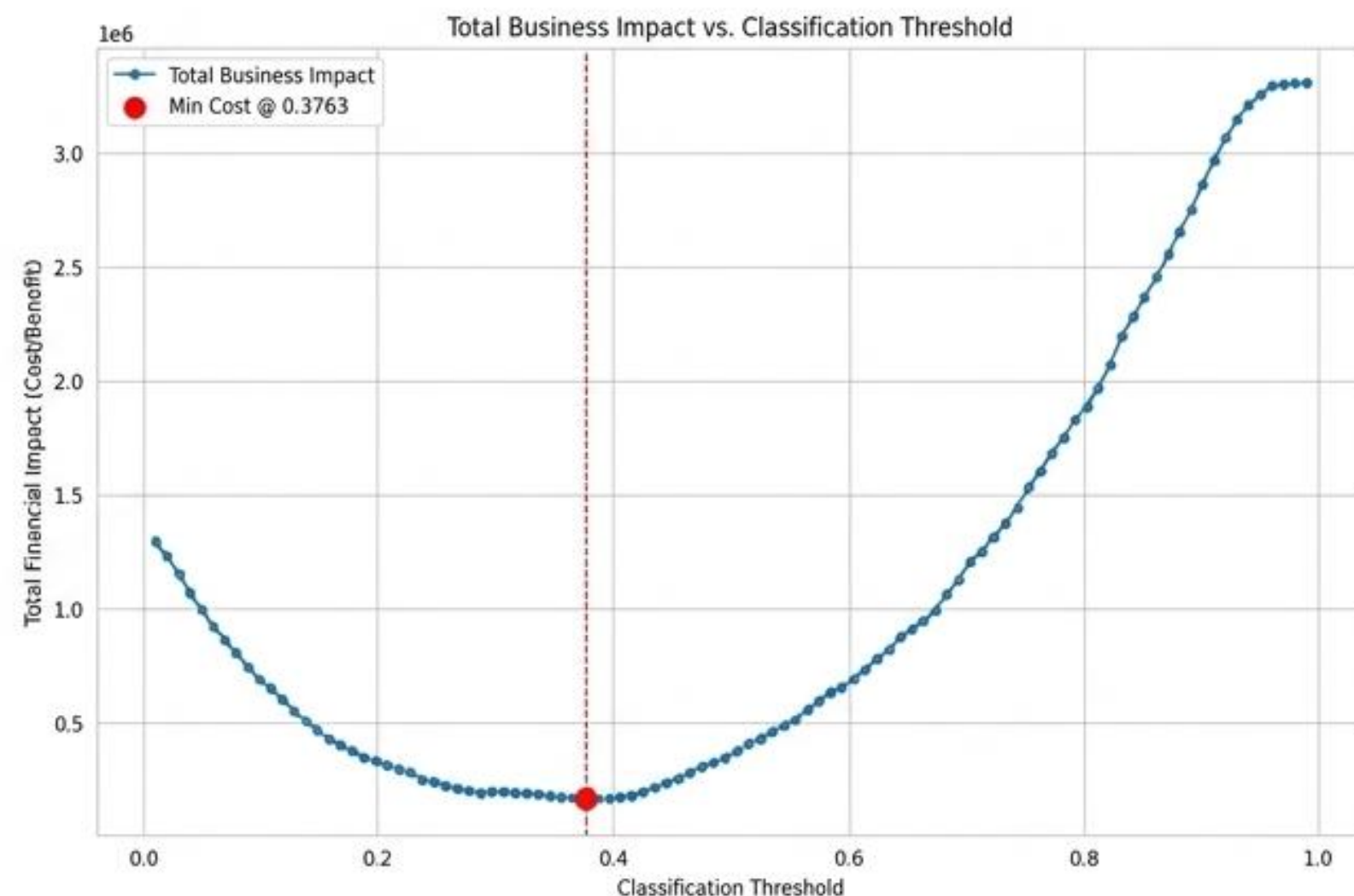


False Positive (False Alarm):
\$10 cost.



True Positive (Caught Fraud):
\$500 saved.

THE FINANCIAL SWEET SPOT



By shifting the threshold to 0.3763, we minimize total exposure, achieving a minimum total cost of \$165,370.

The Glass Box: Explainability via SHAP

Why SHAP?

LIME offers local, case-by-case insights, but SHAP uses game theory to provide consistent global and local explanations. We **know exactly why** every decision is made.

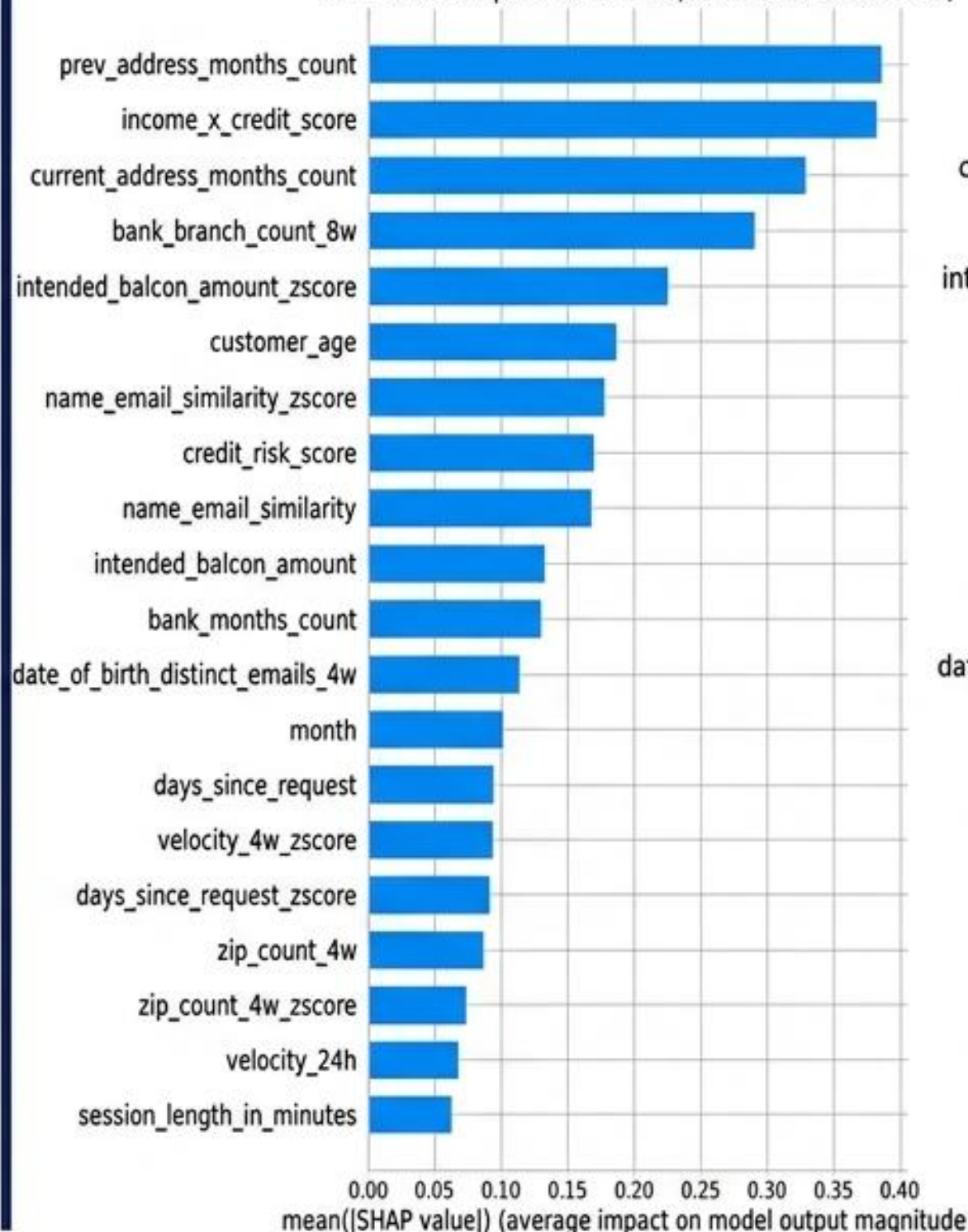
Top Global Drivers

1. prev_address_months_count
(Lower values = High impact on fraud likelihood)

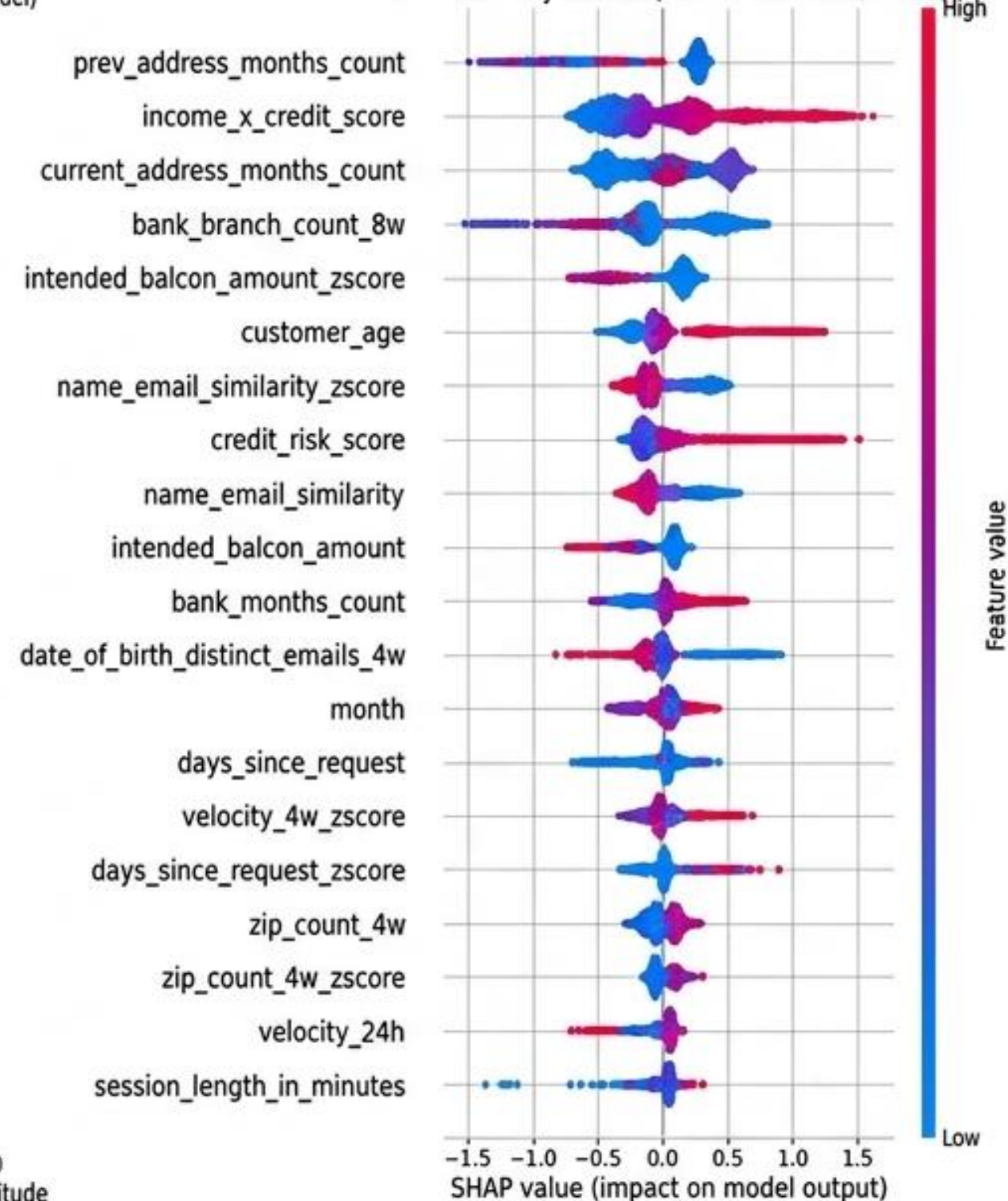
2. income_x_credit_score
(High values = Strong push toward fraud)

3. current_address_months_count
(Recent moves correlate heavily with risk)

SHAP Feature Importance Bar Plot (Fraud Class - Tuned Model)



SHAP Summary Dot Plot (Fraud Class - Tuned Model)



Uncovering Algorithmic Bias (Fairness Audit)

The Ethical Audit: We tested the optimized LightGBM model across sensitive customer age attributes. The model exhibited severe, hidden bias against younger demographics.



<25 Age Group

Selection Rate: **0.0861**

Low TPR (Missing actual fraud)

High FPR (Excessive false alarms)

Systematically generating false alarms and unfair treatment.

Disparate
Impact Ratio
0.1333

Fails the standard 0.8
four-fifths fairness rule.



65+ Age Group

Selection Rate: **0.6455**

High TPR (Catching fraud)

Low FPR (Accurate targeting)

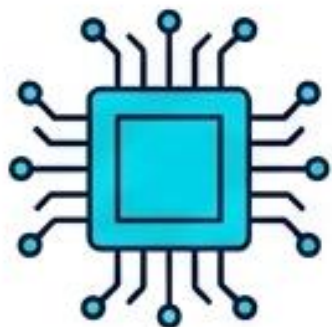
Ethical AI Mitigation Pipeline

Baseline Disparity (Highly Biased) - TPR Difference: 0.455 | FPR Difference: 0.550

Technique	Mechanism	Impact on Equalized Odds
Reweighting	Pre-processing	Marginal impact on disparity.
SMOTE	Data Augmentation	Highly effective. Virtually eliminated FPR gap (0.0004 diff).
Threshold Adjustment	Post-processing	Highly effective. Reduced TPR difference to a negligible 0.003.
Post-processing for Equalized Odds	Direct Optimization	Matched False Positive Rates exactly.

The Result: We successfully debiased the model. SMOTE and custom thresholding ensure equitable treatment across age groups without degrading the \$165K financial optimization.

Deployment Blueprint & Final Summary



Technical Rigor
(The Engine)

AUC
0.832

The Optuna-tuned LightGBM model successfully captures non-linear fraud patterns within highly imbalanced dataset architecture.



Financial ROI
(The Impact)

Threshold
0.376

Custom probability thresholding explicitly minimizes total business loss based on a real-world \$1,000/\$10 cost matrix.



Transparent & Fair
(The Glass Box)

Equalized
Odds

SHAP global explainability guarantees transparency, while active SMOTE bias mitigation ensures equitable treatment across protected age demographics.

STATUS: System audited, debiased, and ready for production deployment.